

APPLICATIONS OF CONGRUENCES



- ❖ Hashing Functions(To assign memory locations to computer files)
- ❖ Pseudorandom Numbers
- ❖ Check Digits

Learning Objective

- ❖ To understand the common technique for detecting errors in digit strings
- ❖ To understand how memory location can be assigned so that customer records can be retrieved quickly

Hashing Functions

- ❖ The central computer at an insurance company maintains records for each of its customers.
- ❖ How can memory locations be assigned so that customer records can be retrieved quickly?
- ❖ The solution is to use a suitably chosen *hashing function*.
- ❖ Records are identified using a key, which usually identifies each customer's records.
- ❖ A hashing function h assigns memory location $h(k)$ to the record that has k as its key.

Hashing Functions(Cont'd)

- ❖ In practice, many different hashing functions are used.
- ❖ One of the most common is the function

$h(k) = k \bmod m$, where m is the number of available memory locations.

Hashing Function(Cont'd)

Example Find the memory locations assigned by the hashing function $h(k) = k \bmod 111$ to the records of customers with Social Security numbers 06421284 and 037149212.

only works for
integers

Solution

$$h(06421284) = 06421284 \bmod 111 = 14.$$

$$h(037149212) = 037149212 \bmod 111 = 65.$$

The record of the customer with social security number 06421284 is assigned to memory location 14.

The record of the customer with social security number 037149212 is assigned to memory location 65.

Pseudorandom Numbers

- ❖ Randomly chosen numbers are often needed for computer simulations.
- ❖ Different methods have been devised for generating numbers that have properties of randomly chosen numbers.
- ❖ Because numbers generated by systematic methods are not truly random, they are called pseudorandom numbers.
- ❖ The most commonly used procedure for generating pseudorandom numbers is the **linear congruential method**.
- ❖ Choose four integers: the **modulus** m , **multiplier** a , **increment** c , and seed x_0 , with $2 \leq a \leq m$, $0 \leq c \leq m$, and $0 \leq x_0 < m$.
- ❖ $x_{n+1} = (a x_n + c) \bmod m$

Pseudorandom Numbers(Cont'd)

Example Find the sequence of pseudorandom numbers generated by the linear congruential method with modulus $m = 9$, multiplier $a = 7$, increment $c = 4$, and seed $x_0 = 3$.

Solution

$$x_{n+1} = (7x_n + 4) \bmod 9$$

$$x_1 = 7x_0 + 4 \bmod 9 = 7 \cdot 3 + 4 \bmod 9 = 25 \bmod 9 = 7,$$

$$x_2 = 7x_1 + 4 \bmod 9 = 7 \cdot 7 + 4 \bmod 9 = 53 \bmod 9 = 8,$$

$$x_3 = 7x_2 + 4 \bmod 9 = 7 \cdot 8 + 4 \bmod 9 = 60 \bmod 9 = 6,$$

$$x_4 = 7x_3 + 4 \bmod 9 = 7 \cdot 6 + 4 \bmod 9 = 46 \bmod 9 = 1,$$

$$x_5 = 7x_4 + 4 \bmod 9 = 7 \cdot 1 + 4 \bmod 9 = 11 \bmod 9 = 2,$$

$$x_6 = 7x_5 + 4 \bmod 9 = 7 \cdot 2 + 4 \bmod 9 = 18 \bmod 9 = 0,$$

Pseudorandom Numbers(Cont'd)

$$x_7 = 7x_6 + 4 \bmod 9 = 7 \cdot 0 + 4 \bmod 9 = 4 \bmod 9 = 4,$$

$$x_8 = 7x_7 + 4 \bmod 9 = 7 \cdot 4 + 4 \bmod 9 = 32 \bmod 9 = 5,$$

$$\swarrow x_9 = 7x_8 + 4 \bmod 9 = 7 \cdot 5 + 4 \bmod 9 = 39 \bmod 9 = 3, \quad \text{—}$$

Since each term depends only on the previous term, the sequence

3, 7, 8, 6, 1, 2, 0, 4, 5, 3, 7, 8, 6, 1, 2, 0, 4, 5, 3, ... is generated.

This sequence contains nine different numbers before repeating.

Check Digits

- ❖ Congruences are used to check for errors in digit strings.
- ❖ A common technique for detecting errors in such strings is to add an extra digit at the end of the string.
- ❖ This final digit, or checking digit, is calculated using a particular function.
- ❖ To determine whether a digit string is correct, a check is made to see whether this final digit has the correct value.

APPLICATIONS OF CONGRUENCES



❖ Example All books are identified by an **International Standard Book Number (ISBN-10)**, a 10 digit code $x_1, x_2, \dots, x_{10}$, assigned by the publisher. An ISBN-10 consists of blocks identifying the language, the publisher, the number assigned to the book by its publishing company, and finally, a check digit that is either a digit or the letter X(used to represent 10). This check digit is selected so that

$$x_{10} \equiv \sum_{i=1}^9 ix_i \pmod{11}, \text{ or equivalently, so that } \sum_{i=1}^{10} ix_i \equiv 0 \pmod{11}.$$

Answer these questions about ISBN-10s:

- (a) The first nine digits of the ISBN-10 of the sixth edition of this book are 007288008. What is the check digit?
- (b) Is 084930149X a valid ISBN-10?

APPLICATIONS OF CONGRUENCES



Solution

(a) Inserting the digits 007288008 gives

$$x_{10} \equiv 1.0 + 2.0 + 3.7 + 4.2 + 5.8 + 6.8 + 7.0 + 8.0 + 9.8 \pmod{11}$$

divided by 11

$$x_{10} \equiv 0 + 0 + 21 + 8 + 40 + 48 + 0 + 0 + 72 \pmod{11}$$

$$x_{10} \equiv 189 \equiv 2 \pmod{11}$$

Hence, $x_{10} = 2$.

(b) To see whether 084930149X is a valid ISBN-10,

$$1.0 + 2.8 + 3.4 + 4.9 + 5.3 + 6.0 + 7.1 + 8.4 + 9.9 + 10.10 = 299$$

$$299 \equiv 2 \not\equiv 0 \pmod{11}.$$

Hence, 084930149X is not a valid ISBN-10.

Summary

- ❖ Hashing Functions
- ❖ Pseudorandom Numbers
- ❖ Check Digits



Example Find the memory locations assigned by the hashing function $h(k) = k \bmod 97$ to the records of insurance company customers with Social Security number 220195744.

Answer: 21

APPLICATIONS OF CONGRUENCES

Next Lecture

Cryptography



APPLICATIONS OF CONGRUENCES

